Agentic RAG:

Tool Routing Under Prompt Ambiguity

Henil Patel · Parin Shah · Heta Shah

Supervised By: Prof. Ryan Rad

CS 6180 · Foundations of Generative AI · Team 8

Project Introduction & Objectives

- **The Routing Decision:** RAG enhances LLM reliability by grounding responses in retrieved documents, but modern agentic systems must dynamically choose between a local knowledge store and live web search at query time
- **Wrong routing = wrong answer:** sending a time-sensitive query to a static document store returns stale data; sending a document-specific query to web search returns irrelevant results
- **Cost of Routing Errors:** Routing errors cause responses that are outdated, off-topic, or confidently wrong, yet most benchmarks evaluate final answer quality, not the routing decision itself
- **The Research Gap:** Despite rapid adoption of agentic RAG in enterprise systems, no principled benchmarks exist that measure how routing accuracy degrades as query ambiguity increases
- **Our Approach:** This project isolates routing as the primary experimental variable, evaluating 6 strategies across 4 LLMs and 3 ambiguity tiers to identify where and why routers fail

Research Question

“ How does routing accuracy degrade as prompt ambiguity increases? ”

Methodology

Routing Strategies

- Rule-Based: temporal keyword matching
- Always-Local: naive baseline (52% ceiling)
- Zero-Shot LLM: direct prompt → JSON
- Few-Shot LLM: 3 labeled examples in context
- CoT LLM: chain-of-thought before classifying
- LangGraph Agent: multi-node stateful workflow

Models Evaluated

- Llama-3-8B-Instruct
- GPT-4o-mini
- Gemma-3-27B-IT
- Claude Sonnet 4.6

Stack & Dataset

- LangChain + LangGraph
- ChromaDB (22 FY2025 10-Ks)
- Tavily real-time web search
- all-MiniLM-L6-v2 embeddings
- 100 queries, 3 ambiguity tiers
- 52 local, 48 web labels

Methodology

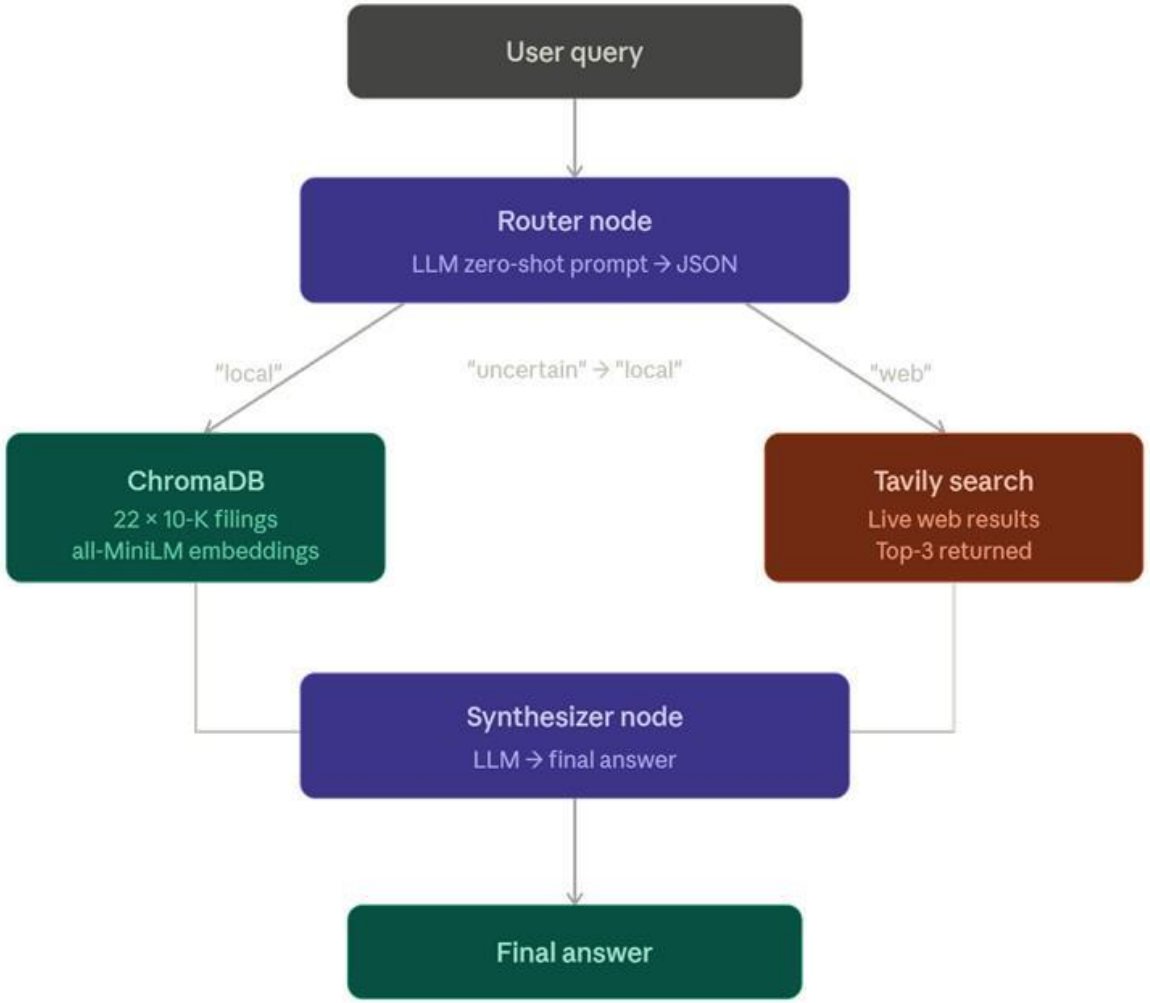
Dataset & Ambiguity Tiers

100 labeled queries across 22 company 10-Ks

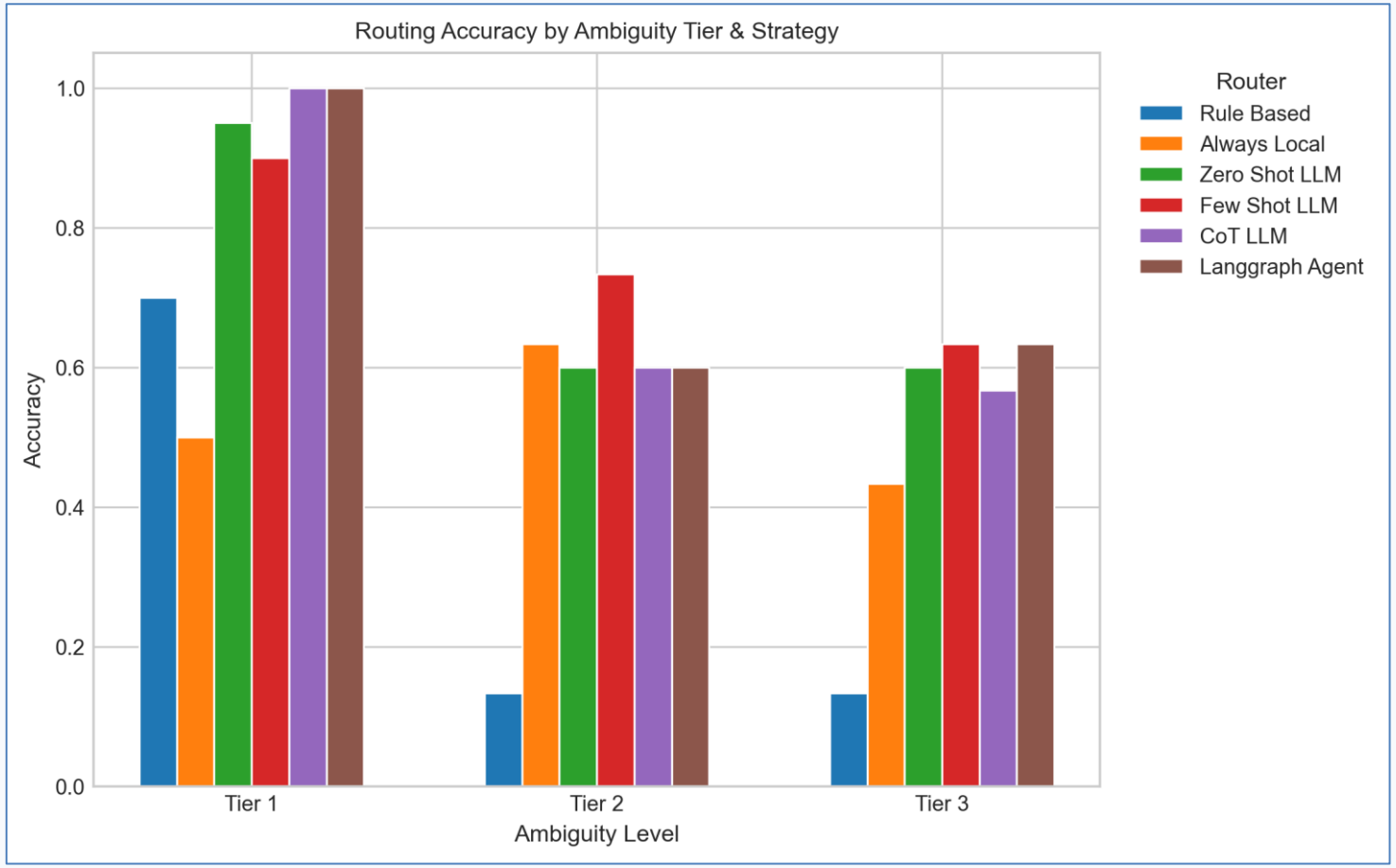
Tier 1 — Clear Intent *(40 queries)* Unambiguous, explicit source signal *"How many employees did Amazon report as of Dec 31, 2025?"* → local *"What is Amazon's stock price right now?"* → web

Tier 2 — Moderately Ambiguous *(30 queries)* Requires inference about recency or scope *"What was Adobe's total revenue for FY2025?"* → local or web *"How is Alphabet integrating Gemini AI?"* → web

Tier 3 — Highly Ambiguous *(30 queries)* Vague or open-ended, no clear source signal *"What is Alphabet's moonshots strategy?" "How is Amazon's growth?"*



Results — Routing Accuracy by Ambiguity Tier

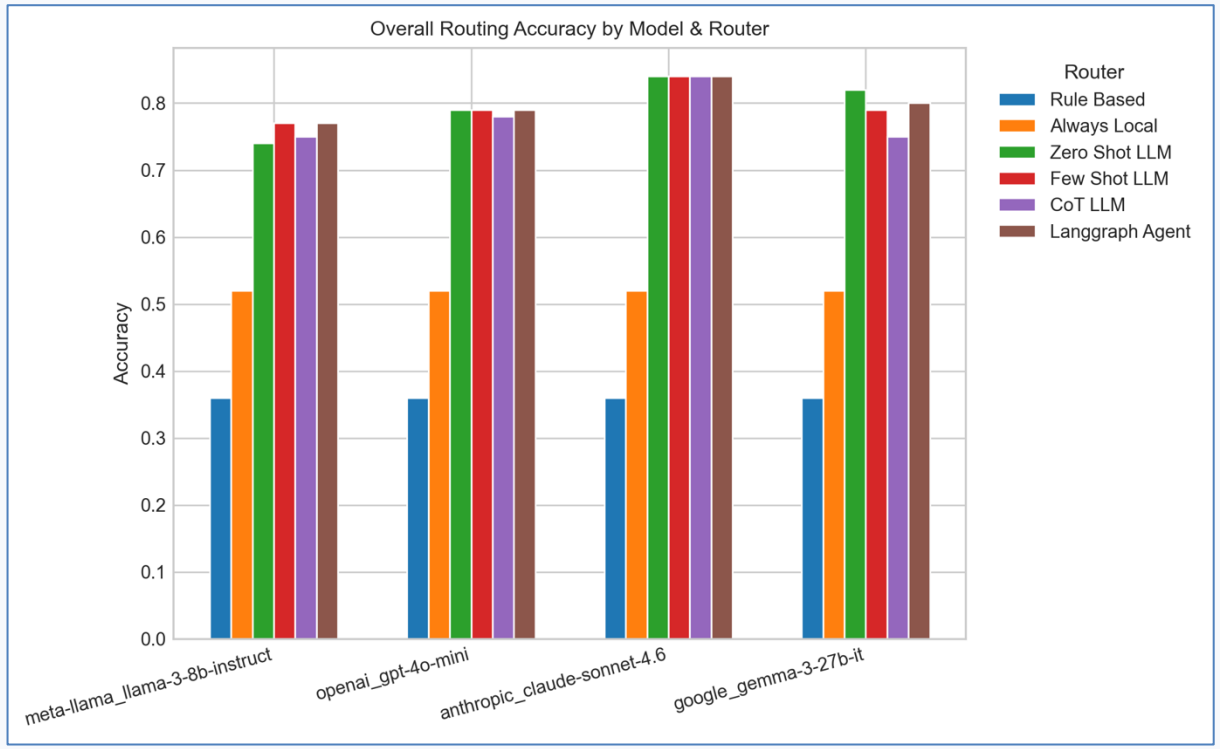


Sonnet 4.6 achieves the highest accuracy (84%) across all LLM-based strategies, followed by Gemma-3-27B (82%), GPT-4o-mini (79%), and LLaMA-3-8B (76%). Rule-based accuracy remains flat at 36% across all models.

All strategies degrade sharply from Tier 1 to Tier 2, with zero-shot dropping from 95% to 60% and few-shot from 90% to 73%. Rule-based collapses entirely at Tiers 2 and 3 (13%).

Model capability matters more than prompt strategy. Claude achieves 84% across all 3 prompt versions, adding few-shot or CoT changes nothing for strong models.

Model Comparison & Failure Patterns



Model Summary

Model	Overall	Tier 1	Tier 3
Claude Sonnet 4.6	84%	100%	63–73%
Gemma 3 27B	75–82%	90–98%	57–67%
GPT 4o mini	78–79%	98–100%	60–67%
Llama 3 8B	74–77%	90–100%	57–63%

Ambiguity is the Ceiling

LLM-based routers achieve high accuracy on clear queries but routing accuracy degrades significantly as query ambiguity increases, dropping from 100% (Tier 1) to 73% (Tier 3) even for the strongest model (Sonnet 4.6). This demonstrates that query ambiguity, not model size or prompt strategy, is the primary bottleneck in agentic RAG routing.

Top Failure Modes

- **Vague-Query Bias**

  Open-ended queries default to web even when the local 10-K has the answer

- **Local → Web Misroutes**

  Risk factors, KPIs, strategy queries sent to web search

- **Temporal Confusion**

  'Early 2026' — annual filing vs. live market data ambiguity

Next Steps

Linguistic Failure Analysis

Examine which specific query features drive mis-routes:

- Missing temporal markers
- Bare entity references (no verb, no scope)
- Underspecified or open-ended queries

Goal: Produce interpretable findings about what makes a query hard to route.

Final Report

Full write-up covering:

- Related work & background
- System design & methodology
- Full experimental results
- Limitations & future work

Thank You